

Reasoning: Syllogism (Complete Study Guide)

Theory, 12 Solved Examples, and 25 Practice Questions with Answer Key

Part 1: Theory (Concepts of Syllogism)

1. What is Syllogism?

Syllogism is a form of logical reasoning where a conclusion is drawn from two or more given statements (premises). In competitive exams, statements are assumed to be true even if they contradict general facts. You have to determine which conclusion(s) logically follow from the given statements.

2. Types of Statements

- **Universal Affirmative (A-type - 'All A are B'):** Represented as circle A completely inside circle B.
- **Universal Negative (E-type - 'No A is B'):** Represented as completely separate circles A and B.
- **Particular Affirmative (I-type - 'Some A are B'):** Represented as overlapping circles A and B.
- **Particular Negative (O-type - 'Some A are not B'):** Part of circle A is outside circle B.

3. Standard Options Format in Syllogism Exams

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Either I or II follows / Both follow (depending on question format)
- (D) Neither I nor II follows

Part 2: Solved Examples (12 Examples)

Example 1

Statements: *All pens are books. All books are pencils.*

Conclusions: I. All pens are pencils. II. Some pencils are books.

Solution: Since all pens are inside books, and all books are inside pencils, all pens are definitely pencils (Conclusion I follows). Also, since all books are pencils, some pencils are books is also true (Conclusion II follows).

Answer: Both conclusions I and II follow.

Example 2

Statements: *Some cats are dogs. All dogs are animals.*

Conclusions: I. Some cats are animals. II. All cats are animals.

Solution: The portion of cats that are dogs will also be animals, so "Some cats are animals" is true (Conclusion I follows). "All cats are animals" is not certain because only some cats are dogs.

Answer: Only conclusion I follows.

Example 3

Statements: *No car is a bus. All bikes are buses.*

Conclusions: I. No bike is a car. II. Some buses are bikes.

Solution: Since no bus is a car and all bikes are buses, bikes can never be cars (Conclusion I follows). Since all bikes are buses, portion of buses is bikes (Conclusion II follows).

Answer: Both conclusions I and II follow.

Example 4

Statements: *All mangoes are fruits. Some fruits are sweet.*

Conclusions: I. Some mangoes are sweet. II. Some sweet are fruits.

Solution: Mangoes are inside fruits, but the 'sweet' overlap might or might not touch mangoes directly (possibility, not certainty). Thus I does not follow. Since some fruits are sweet, reversing it means some sweet are fruits is true.

Answer: Only conclusion II follows.

Example 5

Statements: *All cups are glasses. No glass is a spoon.*

Conclusions: I. No cup is a spoon. II. Some glasses are cups.

Solution: Cups are inside glasses, and glasses have no relation with spoons, meaning cups also have no relation with spoons (I follows). All cups are glasses implies some glasses are cups (II follows).

Answer: Both conclusions I and II follow.

Example 6

Statements: *Some markers are pens. Some pens are markers (re-statement or general). Let's use: Some pens are markers. Some markers are scales.*

Conclusions: I. Some pens are scales. II. No pen is a scale.

Solution: Direct relation between pens and scales is not given, leaving a possibility. However, I and II form a complementary pair (Some + No) for the same elements.

Answer: Either conclusion I or II follows.

Example 7

Statements: *All tables are chairs. Some chairs are wood.*

Conclusions: I. Some tables are wood. II. Some wood are chairs.

Solution: Wood intersects chairs, but may or may not touch tables (I is uncertain). "Some chairs are wood" directly implies "Some wood are chairs" (II follows).

Answer: Only conclusion II follows.

Example 8

Statements: *No flower is a fruit. All fruits are trees.*

Conclusions: I. Some trees are fruits. II. Some trees are not flowers.

Solution: All fruits are trees means some trees are fruits (I follows). Since all fruits are trees and no flower is a fruit, the portion of trees that are fruits can never be flowers (II follows).

Answer: Both conclusions I and II follow.

Example 9

Statements: *All stars are moons. All moons are suns.*

Conclusions: I. All stars are suns. II. Some suns are stars.

Solution: Stars are inside moons, moons are inside suns, so stars are inside suns (I follows). Since all stars are suns, some suns are definitely stars (II follows).

Answer: Both conclusions I and II follow.

Example 10

Statements: *Some papers are books. No book is a magazine.*

Conclusions: I. Some papers are not magazines. II. All magazines are papers.

Solution: The portion of papers that are books can never be magazines (I follows). All magazines are papers is not certain.

Answer: Only conclusion I follows.

Example 11

Statements: *All actors are singers. All singers dancers. (All singers are dancers).*

Conclusions: I. All actors are dancers. II. Some dancers are actors.

Solution: Actors inside singers, singers inside dancers -> Actors inside dancers (I follows). Since all actors are dancers, some dancers are actors (II follows).

Answer: Both conclusions I and II follow.

Example 12

Statements: *Some phones are smart. All smart are electronic.*

Conclusions: I. Some phones are electronic. II. All phones are electronic.

Solution: The portion of phones that are smart will also be electronic (I follows). All phones are electronic is not certain because only some phones are smart.

Answer: Only conclusion I follows.

Part 3: Practice Questions (25 Questions)

Q1.

Statements: All locks are keys. All keys are doors.

Conclusions: I. All locks are doors. II. Some doors are keys.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q2.

Statements: Some pens are pencils. Some pencils are erasers.

Conclusions: I. Some pens are erasers. II. No pen is an eraser.

- (A) Only I follows (B) Only II follows (C) Either I or II (D) Neither follows

Q3.

Statements: All roads are rivers. All rivers are jungles.

Conclusions: I. All roads are jungles. II. Some jungles are rivers.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q4.

Statements: No man is a sky. All skies are birds.

Conclusions: I. No man is a bird. II. Some birds are skies.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q5.

Statements: Some cars are trains. All trains are tracks.

Conclusions: I. Some cars are tracks. II. All cars are tracks.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q6.

Statements: All books are papers. No paper is a pen.

Conclusions: I. No book is a pen. II. Some papers are books.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q7.

Statements: Some cups are plates. Some plates are spoons.

Conclusions: I. Some cups are spoons. II. No cup is a spoon.

- (A) Only I follows (B) Only II follows (C) Either I or II (D) Neither follows

Q8.

Statements: All phones are smart. All smart are fast.

Conclusions: I. All phones are fast. II. Some fast are phones.

(A) Only I follows

(B) Only II follows

(C) Both follow

(D) Neither follows

Q9.

Statements: Some tables are chairs. All chairs are wood.

Conclusions: I. Some tables are wood. II. All tables are wood.

(A) Only I follows

(B) Only II follows

(C) Both follow

(D) Neither follows

Q10.

Statements: No cat is a dog. No dog is an animal.

Conclusions: I. No cat is an animal. II. Some animals are dogs.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows (Wait, let's see II: Some animals are dogs -> Statement says No dog is animal, so II is false. I is false too. So Neither).

Q11.

Statements: All flowers are fruits. Some fruits are sweet.

Conclusions: I. Some flowers are sweet. II. Some sweet are fruits.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q12.

Statements: All pens are markers. All markers are inks.

Conclusions: I. All inks are pens. II. Some inks are pens.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q13.

Statements: Some bags are books. All books are heavy.

Conclusions: I. Some bags are heavy. II. All heavy are bags.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q14.

Statements: No star is a planet. All planets are suns.

Conclusions: I. Some suns are planets. II. Some suns are not stars.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q15.

Statements: Some actors are singers. All singers are dancers.

Conclusions: I. Some actors are dancers. II. All actors are dancers.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q16.

Statements: All computers are laptops. No laptop is a TV.

Conclusions: I. No computer is a TV. II. Some laptops are computers.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q17.

Statements: Some birds are animals. Some animals are insects.

Conclusions: I. Some birds are insects. II. No bird is an insect.

- (A) Only I follows (B) Only II follows (C) Either I or II (D) Neither follows

Q18.

Statements: All shirts are pants. All pants are coats.

Conclusions: I. All shirts are coats. II. All coats are shirts.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q19.

Statements: Some stones are rocks. All rocks are mountains.

Conclusions: I. Some stones are mountains. II. All stones are mountains.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q20.

Statements: No tree is a plant. All plants are shrubs.

Conclusions: I. Some shrubs are plants. II. Some shrubs are not trees.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q21.

Statements: All boxes are bags. Some bags are suitcases.

Conclusions: I. Some boxes are suitcases. II. Some suitcases are bags.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q22.

Statements: Some pens are pencils. All pencils are markers.

Conclusions: I. Some pens are markers. II. All markers are pens.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q23.

Statements: All kings are queens. All queens are princes.

Conclusions: I. All kings are princes. II. Some princes are queens.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q24.

Statements: No water is milk. All milk is tea.

Conclusions: I. No water is tea. II. Some tea is milk.

- (A) Only I follows (B) Only II follows (C) Both follow (D) Neither follows

Q25.

Statements: Some doors are windows. Some windows are rooms.

Conclusions: I. Some doors are rooms. II. No door is a room.

(A) Only I follows

(B) Only II follows

(C) Either I or II follows

(D) Neither follows

Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(C) Both follow	Q8	(C) Both follow	Q15	(A) Only I follows	Q22	(A) Only I follows
Q2	(C) Either I or II	Q9	(A) Only I follows	Q16	(C) Both follow	Q23	(C) Both follow
Q3	(C) Both follow	Q10	(D) Neither follows	Q17	(C) Either I or II	Q24	(B) Only II follows
Q4	(B) Only II follows	Q11	(B) Only II follows	Q18	(A) Only I follows	Q25	(C) Either I or II follows
Q5	(A) Only I follows	Q12	(B) Only II follows	Q19	(A) Only I follows	-	-
Q6	(C) Both follow	Q13	(A) Only I follows	Q20	(C) Both follow	-	-
Q7	(C) Either I or II	Q14	(C) Both follow	Q21	(B) Only II follows	-	-